Reg. No. | | | | | | | | | | |

[16CIC401]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2018

DISCRETE MATHEMATICS AND GRAPH THEORY

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Instructions: Answer any **FIVE** full questions.Choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed

PART - A

1. a. In a sample of 100 logic chips, 23 have a defect D_1, 26 have defect D_2, 30 have defect D_3, 7 have defect $D_1$and D_2, 8 have defect D_1 and D_3, 10 have defect D_2 and D_3 and 3 have all the three Defects. Find the number of chips having **07**

i. Atleast one defect

ii. No defect

b. Let H be a function defined for all nonnegative integers by the following recursive definition $H(0)=1$, $H(1)=2$, $H(N+2)=H(N)^2+H(N+1)$, $N\geq 0$.Compute first six values of H. **04**

C. What are characteristic functions? Prove that $f_{A\cup B}=f_A+f_B-f_Af_B$ **03**

2. a. Define tautology? Prove that $[(p\to q)\wedge(q\to r)]\to(p\to r)$ is tautology. **07**

b. Prove that for all integers K and l are both odd then K+l is even and Kl is odd **04**

c. Prove by mathematical induction $1+2+3+.....+n=\frac{n(n+1)}{2}$, $n\geq 1$ **03**

3. a. Solve the recurrence relations **07**

i. $a_n=7a_{n-1}$, $n\geq 1$ with $a_2=98$

ii. $a_n=na_{n-1}, n\geq 1$ with $a_0=1$

b. Prove that if 30 dictionaries in a library contain a total of 61,327 pages, then atleast one of the dictionaries must have atleast 2045 pages. **04**

c. Define: **03**

i. Graph

ii. Loop

iii. Simple path

PART-B

4. a. Let A= $\{1,2,3,4,5\}$ Define a relation R on A$\times A$ by (x_1, y_1)R(x_2, y_2) if and only if $x_1 + y_1 = x_2 + y_2$. **07**

   i. Verify R is an equivalence relation on R

   ii. Determine the equivalence classes [1,3] [2,4] and [1,1]

   iii. Determine the partition on A$\times A$ induced by R.

   b. Consider the relation R= $\{(a,a)(a,b)(b,c)(c,d)(c,e)(d,e)$ defined on the set A $=\{a,b,c,d,e\}$ find M_{R^∞}. **04**

   c. Let $R_1 = \{(1,x)(2,x)(3,y)(3,z)\}$ $R_2 = \{(w,5)(w,6)\}$ on A= $\{1,2,3,4\}$ B= $\{w,x,y,z\}$ and C= $\{5,6,7\}$. Compute $M(R_1)M(R_2)M(R_1 o R_2)$ **03**

PART - B

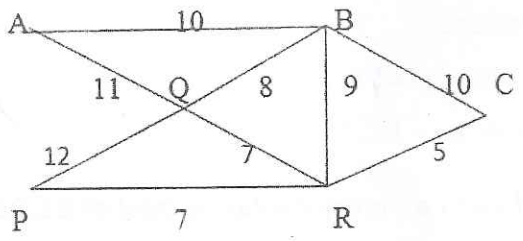
5. a. If R is a relation on the set A= $\{1,2,3,4\}$ defined by xRy such that x/y prove that (A,R) is a poset. Draw the Hasse diagram.

   b. Consider the Boolean polynomial $P=(x,y,z)=x\wedge(y\vee z')$, if B= $\{0,1\}$

   i. Construct truth table for the Boolean function $:B_3 \to B$ defined by Boolean polynomial **04**

   ii. Construct the logic diagram implementing the function f. **03**

6. a. Use Kruskal's algorithm to find minimal spanning tree for the weighted graph shown below. **07**



   b. Prove that A tree with n vertices has $n-1$ edges. **04**

   c. Define: **03**

   i. Binary tree

   ii. Balance tree

   iii. Full binary tree

7. a. Define Abelian group. Let G be the set of all non zero real numbers and let $a * b = \frac{ab}{2}$ Show that $(G,*)$ is an abelian group. **07**

b. Define Semi group and Monoid. If G is group and 'a' and 'b' are elements of the group G, then show that **07**

i. $(a^{-1})^{-1} = a$

ii. $(ab)^{-1} = b^{-1}a^{-1}$

8. a. Explain the following functions with an example

i. Identity function

ii. Constant function

iii. Onto function

iv. One to one function

v. One to one correspondence.

b. Let $A = \{1,2,3,4\}$ and f and g be functions from A to A given by $f = \{(1,4)(2,1)(3,2)(4,3)\}$ $g = \{(1,2)(2,3)(3,4)(4,1)\}$ .Prove that f and g are inverses of each other.